Mohamad Latifian

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Research Interests

My research lies at the intersection of theoretical computer science, artificial intelligence, and economics, with a focus on algorithms and computational complexity. My primary interests include computational social choice, fair division, algorithmic game theory, and mechanism design.

Education and Academic Positions

Postdoctoral Research Associate *Jul. 2024 – present*
University of Edinburgh, UK
Host: *Aris Filos-Ratsikas*

Ph.D. in Computer Science *Sep. 2020 – Sep. 2024*
University of Toronto, Canada
Advisors: *Nisarg Shah and Allan Borodin*
Thesis: Decoding Distortion: Pushing the Limits of Optimal Collective Decision-Making
GPA: *4.0/4.0*

M.Sc. in Algorithms and Computation *Sep. 2017 – Sep. 2019*
Sharif University of Technology, Iran
Advisor: *Mohammad Ghodsi*
Thesis: *Considering the Effect of Abstention in the Distortion Value.*
GPA: *19.59/20*

B.Sc. in Computer Engineering *Sep. 2012 – Jul. 2017*
Sharif University of Technology, Iran
Advisor: *Hamid Zarrabi-Zadeh*
Thesis: *Hamiltonian Path in Solid Grid Graphs.*
GPA: *17.22/20*

Conference Publications

(Authors in alphabetical order)

C19. *Nash Equilibria in Auctions with Pacing Strategies: Complexity and Inefficiency*. A. Filos-Ratsikas, C. Kokkalis, and M. Latifian.
Proc. of the 22nd Conference on Web and Internet Economics (**WINE**), 2026, (Forthcoming).

C18. *Social Deliberation: Democratic Deliberation Through Unbiased Social Interactions*. D. Afazeli, M. Latifian, and M. Seddighin.
Proc. of the ACM Web Conference 2026 (**WWW**), pp. 375–383, 2026.

C17. *Metric Distortion with Preference Intensities*. M. Abbaszadeh, A. Ansarifar, M. Latifian, and M. Seddighin.
Proc. of the 40th AAAI Conference on Artificial Intelligence (**AAAI**), volume 40, pp. 16603–16611, 2026.

C16. *Optimal Metric Distortion for Matching on the Line*. A. Filos-Ratsikas, V. Gkatzelis, M. Latifian, E. Rewinski, and A. A. Voudouris.
Proc. of the 34th International Joint Conference on Artificial Intelligence (**IJCAI**), pp. 3857–3865, 2025.

C15. *Towards Fair and Efficient Public Transportation: A Bus Stop Model*. M. Bullinger, E. Elkind, and M. Latifian.
Proc. of the 24th International Conference on Autonomous Agents and Multiagent Systems (**AAMAS**), pp. 427–435, 2025.

C14. *Temporal Fair Division of Indivisible Items*. E. Elkind, A. Lam, M. Latifian, T. Y. Neoh, and N. Teh.
Proc. of the 24th International Conference on Autonomous Agents and Multiagent Systems (**AAMAS**), pp. 676–685, 2025.

C13. *Fair Division with Market Values*. S. Barman, S. Ebadian, M. Latifian, and N. Shah.
Proc. of the 39th AAAI Conference on Artificial Intelligence (**AAAI**), volume 39, pp. 13589–13596, 2025.

C12. *The Distortion of Public-Spirited Participatory Budgeting*. M. Bedaywi, B. Flanigan, M. Latifian, and N. Shah.
Proc. of the 39th AAAI Conference on Artificial Intelligence (**AAAI**), volume 39, pp. 13605–13613, 2025.

C11. *The Distortion of Threshold Approval Matching*. M. Latifian and A. A. Voudouris.
Proc. of the 33rd International Joint Conference on Artificial Intelligence (**IJCAI**), pp. 2851–2859, 2024.

C10. *Computational Aspects of Distortion*. S. Ebadian, A. Filos-Ratsikas, M. Latifian, and N. Shah.
Proc. of the 23rd International Conference on Autonomous Agents and Multiagent Systems (**AAMAS**), pp. 499–512, 2024.

C9. *Explainable and Efficient Randomized Voting Rules*. S. Ebadian, A. Filos-Ratsikas, M. Latifian, and N. Shah.
Proc. of the 37th Conference on Neural Information Processing Systems (**NeurIPS**), volume 36, 2023.

C8. *Best of Both Distortion Worlds*. V. Gkatzelis, M. Latifian, and N. Shah.
Proc. of the 24th ACM Conference on Economics and Computation (**EC**), pp. 738–758, 2023.

C7. *The Distortion of Approval Voting with Runoff*. S. Ebadian, M. Latifian, and N. Shah.
Proc. of the 22nd International Conference on Autonomous Agents and Multiagent Systems (**AAMAS**), pp. 1752–1760, 2023.

C6. *Voting with Preference Intensities*. A. Kahng, M. Latifian, and N. Shah.
Proc. of the 37th AAAI Conference on Artificial Intelligence (**AAAI**), pp. 5697–5704, 2023.

C5. *Distortion in Voting with Top-t Preferences*. A. Borodin, D. Halpern, M. Latifian, and N. Shah.
Proc. of the 31st International Joint Conference on Artificial Intelligence (**IJCAI**), pp. 116–122, 2022.

C4. *Almost Envy-freeness, Envy-rank, and Nash Social Welfare Matchings*. A. Farhadi, M. Hajiaghayi, M. Latifian, M. Seddighin, and H. Yami.
Proc. of the 35th AAAI Conference on Artificial Intelligence (**AAAI**), pp. 5355–5362, 2021.

C3. *Resource-Aware Protocols for Network Cost-Sharing Games*. G. Christodoulou, V. Gkatzelis, M. Latifian, and A. Sgouritsa.
Proc. of the 21st ACM Conference on Economics and Computation (**EC**), pp. 81–107, 2020.

C2. *On the Distortion Value of the Elections with Abstention*. M. Ghodsi, M. Latifian, and M. Seddighin.
Proc. of the 33rd AAAI Conference on Artificial Intelligence (**AAAI**), pp. 1981–1988, 2019.

C1. *Rent Division Among Groups*. M. Ghodsi, M. Latifian, A. Mohammadi, S. Moradian, and M. Seddighin.
Proc. of the 12th International Conference on Combinatorial Optimization and Applications (**COCOA**), pp. 577–591, 2018.

Journal Publications

J4. *Optimal Metric Distortion for Matching on the Line*. A. Filos-Ratsikas, V. Gkatzelis, M. Latifian, E. Rewinski, and A. A. Voudouris. Artificial Intelligence, 361:104630, 2026.

J3. *Computational Aspects of Distortion*. S. Ebadian, A. Filos-Ratsikas, M. Latifian, and N. Shah. Autonomous Agents and Multi-Agent Systems, (Forthcoming).

J2. *The Distortion of Threshold Approval Matching*. M. Latifian and A. A. Voudouris. Autonomous Agents and Multi-Agent Systems, 39(2):42, 2025.

J1. *On the Distortion Value of Elections with Abstention*. M. Ghodsi, M. Latifian, and M. Seddighin. Journal of Artificial Intelligence Research, 70:567–595, 2021.

Working Papers

(Authors in alphabetical order)

M6. *The Distortion of Threshold Approval Multiwinner Voting*.
M. Latifian, M. Nilipour, and A. A. Voudouris, 2026, arXiv:2610.13725. Under submission to AAMAS '27.

M5. *Group Assignment: Bridging Voting and Matching through Distortion*.
M. Latifian.

M4. *Collective Decision Making with a Few Coin Flips*.
M. Latifian. Under submission to AAMAS '27.

M3. *Envy-Free Cake-Cutting for Five Agents*.
A. Filos-Ratsikas, A. Hollender, M. Latifian, and G. Tyrovolas.

M2. *Metric Distortion with Graph Objectives*.
A. Filos-Ratsikas, M. H. Ghasemi, M. Latifian, and M. Sharify. Under submission to AAMAS '27.

M1. *Distortion of Truthful Matching Mechanisms*.
V. Gkatzelis, M. Latifian, and E. Rewinski.

Academic Visits

Long-Term

University of Oxford, UK. *Sep. – Nov. 2023*
Host: Edith Elkind

Max Planck Institute for Informatics, Germany. *May – Jun. 2019*
Host: Alkmini Sgouritsa
Work title: *Resource-Aware Protocols for Network Cost-Sharing Games* (C3.)

Short-Term

University of Oxford, UK. *Sep. 2025*
Host: Alexandros Hollender

University of Edinburgh, UK. *Nov. 2023*
Host: Aris Filos-Ratsikas

University of Essex, UK. *Oct. 2023*
Host: Alexandros Voudouris

Professional Service

- **PC Member:** AAAI '27, EC '26, AAAI '26, EC '25, AAMAS '25, ICLR '25, AAAI '25, NeurIPS '24, IJCAI '24, AAMAS '24, AAAI '24, AAAI '21
- **Journal Reviewer:** Autonomous Agents and Multi-Agent Systems (JAAMAS), Artificial Intelligence (AIJ), Journal of Artificial Intelligence Research (JAIR)
- **Conference Reviewer:** SODA '27, ESA '26, STOC '25

Invited Talks

T7. *Fair Division with Market Values*, Workshop on Fair Resource Allocation: Concepts, Algorithms and Complexity (FaiRACAC '25), held in conjunction with ICALP 2025 (7 Jul. 2025).

T6. *Distortion in Social Choice Problems*, University of Edinburgh, EconCS Group Meetings (Oct. 2024).

T5. *Fair Division with Market Values*, University of Edinburgh, Laboratory for Foundations of Computer Science (LFCS) Lab Lunch (Oct. 2024).

T4. *How to Make Efficient (Collective) Decisions*, Tehran Institute for Advanced Studies (Sep. 2024).

T3. *Best of Both Distortion Worlds*, University of Edinburgh, Laboratory for Foundations of Computer Science (LFCS) Seminar (Nov. 2023).

T2. *Best of Both Distortion Worlds*, University of Essex, The Centre for Computational Finance and Economic Agents (CCFEA) Seminar Series (Oct. 2023)

T1. *Resource-Aware Protocols for Network Cost-Sharing Games*, The Drexel EconCS Research Group Meetings (Jun. 2020).

Selected Teaching Experience

Guest Lecturer *Oct. 2025*
University of Edinburgh
Algorithms and Data Structures (*Course web page*)

Teaching Assistant
University of Toronto

- **Algorithm Design, Analysis & Complexity** Fall '23
- **Data Structures and Analysis** Winter '23
- **Algorithmic Game Theory and Mechanism Design** Fall '22

Teaching Assistant (* := Lead TA)
Sharif University of Technology

- **Algorithmic Game Theory** Fall '17, Spring '19*
- **Design and Analysis of Algorithms** Fall '14, Fall '18*
- **Theory of Languages and Automata** Fall '14, Spring '15, Fall '15, Spring '16
- **Data Structures and Algorithms** Spring '15

Member of the Scientific Committee *Fall 2016*
4th Sharif National Students Competition,
Sharif University of Technology.

Head of the Scientific Committee *Aug. 2013 and Aug. 2014*
13th & 14th Mehregan Mathematics Workshop,
Shahid Ejeii High School.

Teaching Special Topics in Mathematics *2011–2013*
Teaching high-school students preparing for Olympiad in Informatics.
Topics included Algorithms, Graph Theory, Problem-Solving Strategies, and Programming.

Supervision Experience

Mohammad Hossein Ghasemi and Mohammad Sharify (*Co-supervised with A. Filos-Ratsikas*)
Undergraduate students, Sharif University of Technology *Aug. 2025 – present*
Title: *Metric Distortion with Graph Objectives*

Hamidreza Alipour, Kiarash Joolaei, Sepehr Mizanian
Undergraduate students, Sharif University of Technology *Nov. 2024 – present*
Title: *A Deeper Look at Distortion in Impartial Culture Electorates*

Dana Afazeli
Undergraduate student, Sharif University of Technology *Aug. 2024 – Jan. 2026*
Title: *Social Deliberation: Democratic Deliberation Through Unbiased Social Interactions* (*Appeared in WWW '26*)

Mehrad Abbaszadeh and Ali Ansarifar (*Co-supervised with M. Seddighin*)
Undergraduate students, Sharif University of Technology *Oct. 2023 – Aug. 2025*
Title: *Metric Distortion with Preference Intensities* (*Appeared in AAAI '26*)

Honors and Awards

- **C.C. Gotlieb (Kelly) Graduate Fellowship** in the Department of Computer Science, University of Toronto *2023*
- **Silver medal** in the 21st Iranian National Olympiad in Informatics (INOI). *2011*
- **Ranked 2nd and 7th** in Nationwide Graduate Entrance Exam, in Algorithms and Computation and Artificial Intelligence majors, respectively. *2017*
- **Ranked 1st** in cumulative GPA among 97 M.Sc. students of the department, Sharif University of Technology. *2019*
- Scholarship from Iranian National Elites Foundation *2012 – 2019*

Activities

- **Elected Member & President** of the Students' Scientific Chapter (SSC) Computer Engineering Department of Sharif University of Technology. *2014 – 2015*
- **Chief of Staff** of 17th & 18th International Collegiate Programming Contest (ICPC) Asia Region - Tehran Site. *2015 – 2016*
- **Chief of Staff** of 6th & 7th AI Challenge *2014 – 2015*
  An event where AI agents written in Java, C++, or Python compete to pass the game field and overcome other agents.